From force diagrams to motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a force diagram to predict acceleration and velocity change.

SUCCESS CRITERIA - I can...

1. Write a force sum along each chosen axis.
2. Connect net force and mass to acceleration.
3. Use velocity direction to decide the initial speed change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **3 kg** cart has **5 N** right and **26 N** left.
1. **Find acceleration**.
2. **What else** must you know to decide if it speeds up?

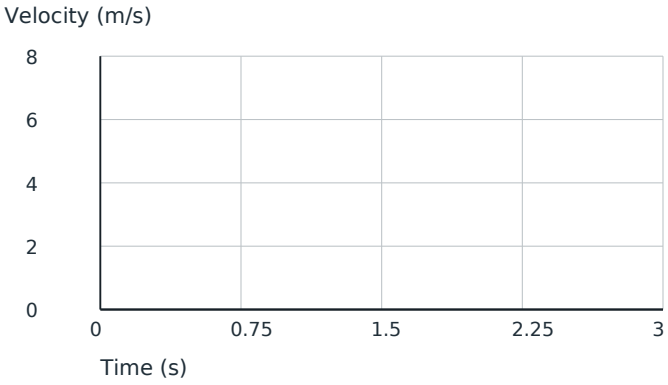
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F_{g,\ parallel} = mg \sin\theta$; $F_{g,\ perpendicular} = mg \cos\theta$
$\Sigma F = ma$.
Ramp angle θ above horizontal. Components replace the weight vector; they are not extra forces.

Worked reasoning: A **2 kg** cart initially at rest has a constant **4 N** net force right for **3 s**. Acceleration is **2 m/s²**; final speed is **6 m/s** right.

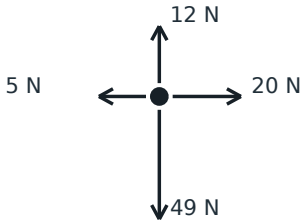
Your turn: Draw a force diagram and a velocity-time graph that agree. **Mark where mass enters the** reasoning.



Marzano check - circle: ① ② ③

03 Angled pull

A **5 kg** box stays on a level floor. A rope pulls with components **20 N** right and **12 N** up; friction is **5 N** left.
1. **Find normal force** and horizontal acceleration.
2. Use g=**9.8**.

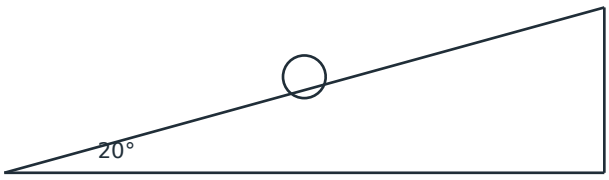


The dot represents the object. Normal not shown; find it

Marzano check - circle: ① ② ③

04 Frictionless ramp

A **2 kg** cart slides down a **20°** frictionless ramp.
1. **Find normal force** and acceleration along the ramp.
2. Use g=**9.8 m/s²**.



The marker represents the object on the sloping surface.

Marzano check - circle: ① ② ③

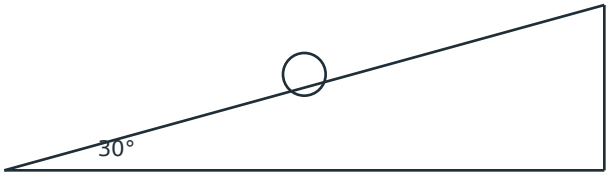
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Continue down each column.

05 Oppose upward motion

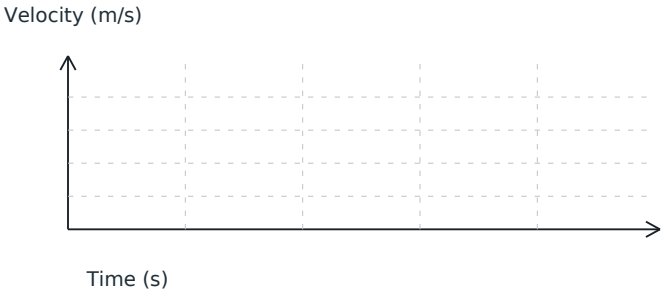
A **4 kg** block slides up a **30°** ramp after release. Kinetic friction is **5 N** down the ramp. Use g = **9.8 m/s²** and **choose up the ramp as** positive. **Complete the force diagram**, calculate acceleration, and **describe the initial change** in speed.



Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

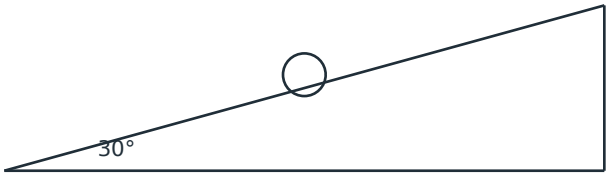
Velocity is **+5 m/s** for **7 s**, then **-4 m/s** for **5 s**. **Find displacement** and distance. **Explain the reference information needed** for an angle of **35°**.
Sketch the velocity–time graph; **label the given values**.



Marzano check - circle: 1 2 3

07 Exit check

A **2 kg** block slides down a **30°** ramp with **3.8 N** friction up the ramp.
1. **Find along-ramp acceleration** and normal force.
2. Use g=**9.8**.

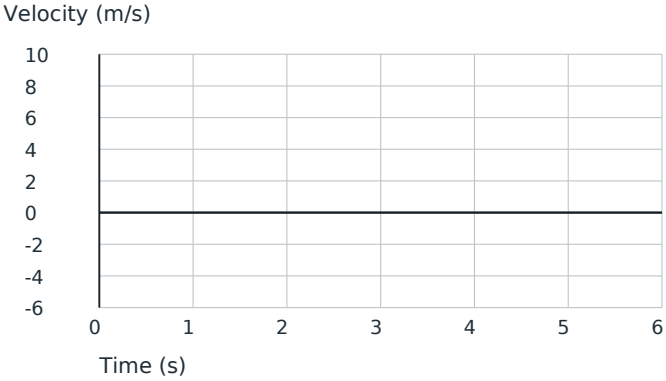


2 kg block slides down; friction 3.8 N up the ramp.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **2 kg** cart moving right at **8 m/s** has a constant **4 N** net force left for **6 s**. Find when it turns, final velocity, displacement and distance.



Marzano check - circle: 1 2 3